ReSkip: Adaptive Depth for Pretrained Transformers via Attention Residuals

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Abstract

Pretrained transformers execute every layer for every input. Adding input-dependent depth allocation typically requires either a separate router / exit head, or rebuilding the residual structure from scratch. We study a third option: installing an intrinsic depth-routing signal into a standard pretrained transformer through a short fine-tune. Building on Attention Residuals (ATTNRES) [24], whose softmax routing over previous block outputs is computed *before* each block runs, our main contribution is AR-RETROFIT, an identity-preserving γ-gated residual-injection fine-tune that freezes the base and adds well under 1% new parameters. On Qwen3-VL-2B and 4B, the retrofit improves the base on 5 of 6 `lmms-eval` VLM benchmarks at 2B (one cell flat) and all 6 at 4B, at near-parity inference cost with retrofit + dynamic RESKIP composed ($\approx 1.02\times$ base wall-clock). A LIBERO VLA policy warm-started from the retrofit improves a matched OFT baseline at both 2B and 4B. We further show that the installed routing weights can be calibrated, with a complementary safety check, into a dynamic block-skip rule (RESKIP) with no auxiliary head, and that the same signal supports adaptive depth on the action stream of the trained policy. The result is a practical mechanism for installing depth-axis adaptive computation into already-trained transformers, rather than retraining the residual structure from scratch.

1 Introduction

Different inputs should not always spend the same computation. The dual-process distinction [18] contrasts *System 1* (fast, automatic) with *System 2* (slow, deliberate); the same dichotomy has direct correlates in cortical processing, where easy stimuli are recognised through shallow feedforward sweeps and harder ones recruit recurrent and deeper circuits [25, 19, 21, 42]. Both perspectives suggest the same computational target: *the agent decides its own depth, per input*. We use this analogy only as a design motivation, not a cognitive-faithfulness claim. The technical question is whether depth allocation can be exposed as an *intrinsic* signal of the transformer forward pass, rather than delegated to a separate router or exit head.

Large language models already expose one System-2-like axis: *tokens*. Test-time compute scaling [34, 8, 41], chain-of-thought prompting [45], and self-thought methods [48] spend more compute on hard problems by emitting more reasoning tokens before answering. Yet every token in such a trace still traverses every layer of the underlying transformer, regardless of whether that token is a routine connector or a load-bearing inferential step. The missing complementary axis is therefore *depth*: deciding which blocks to invoke for this input, at this point in the sequence.

Existing depth-axis methods do not quite match this motivation because they externalise the decision. Early-exit classifiers [40, 12] add auxiliary heads per layer; Mixture-of-Depths [39] learns a token router with a capacity-balancing loss; layer-pruning [16, 32] is post-hoc and static; Universal Transformers [9] share weights with ACT-style halting [15]. In each case a separate component decides whether the rest of the network should run. Cognitive and neural System-2 control, in contrast, is not a separate module; it is a property of the substrate.

ATTNRES [24] gives that intrinsic handle. It replaces the fixed scalar-1 residual with a learned softmax-attention over previous block outputs; the routing weights $\alpha_{i\to n}$ are input-dependent, competitively normalised, and computed *before* block n runs, as part of forming its input. They therefore emit a per-block depth-allocation signal as a side-effect of the network's own forward, not as a separate head. The practical obstacle is that every existing ATTNRES model is pretrained from scratch with the modified residual; reproducing this at the 2–7B scale that real applications need costs tens of thousands of GPU-hours (Appendix D.1). The community has already invested in strong standard-residual transformers—Qwen3-VL, LLaVA-OneVision, InternVL—that no group will retrain just to acquire a depth-axis routing signal. The central problem in this paper is how to install such a signal into a standard pretrained transformer through a short fine-tune, while preserving its original capabilities and making the signal usable for skipping.

The paper makes one main contribution and two supporting ones.

- **AR-RETROFIT** (§2.3, *main*). A γ-gated residual injection installs ATTNRES into a frozen pretrained transformer through a single short fine-tune ($\sim 0.7\%$ new parameters, identity-at-init, tens of GPU-minutes per billion), with every block converging to γ_n=1. On Qwen3-VL-2B/4B the retrofit improves the base on 5 of 6 VLM benchmarks at 2B (one flat) and all 6 at 4B, at near-parity inference cost with dynamic RESKIP (§3).
- **Routing weights as a usable depth signal** (§2.2). Once installed, the pre-execution ATTNRES weights drive a per-input layer-skip rule (RESKIP) with no auxiliary head. The signal is necessary but not sufficient—we additionally require an offline ablation safety check—and at a rate-matched comparison the input-dependent rule strictly beats static and random skip schedules at the same skip rate (Appendix D.5).
- **Transfer to embodied control** (§4). A LIBERO VLA warm-started from the retrofit improves a matched pure-OFT baseline at both 2B and 4B; RESKIP on the action stream, with thresholds re-calibrated on the action distribution, preserves the trained policy at the conservative operating point (+1.10pp on 2B; within seed noise on 4B).

2 Method

2.1 Background: Attention Residuals

A standard transformer uses fixed scalar-1 combination, $\mathbf{x}_n = \mathbf{x}_{n-1} + f_n(\mathbf{x}_{n-1})$, emitting no routing signal. ATTNRES [24] replaces this with a learned attention over depth. Each block n maintains a pseudo-query $\mathbf{w}_n \in \mathbb{R}^d$; keys are projections $\mathbf{k}_i = W_K \mathbf{h}_i$ of earlier block outputs $\mathbf{h}_i$, and

$$\alpha_{i\to n} = \frac{\exp(\mathbf{w}_n^\top \mathbf{k}_i/\sqrt{d})}{\sum_{j=0}^{n-1}\exp(\mathbf{w}_n^\top \mathbf{k}_j/\sqrt{d})}, \qquad \mathbf{x}_n = \sum_{i=0}^{n-1} \alpha_{i\to n}\,\mathbf{h}_i. \tag{1}$$

The key property is two-phase execution. Computing the input to block n requires only $\{\mathbf{h}_i\}_{i<n}$ and $\mathbf{w}_n$—both available *before* f_n runs. We call the pre-execution attention computation *phase 1* and f_n itself *phase 2*. Phase 1 is the routing signal we exploit. Block-ATTNRES groups layers into N blocks and applies the softmax at the block level (the granularity used throughout). Algorithm 1 (Appendix C.7) gives the full two-phase forward with RESKIP's skip check folded in.

2.2 RESKIP: AttnRes-guided layer skipping

RESKIP turns the phase-1 signal into a per-input skip rule. Each block n has two offline scores: an ATTNRES *importance* $I(n) = \max_{l>n} \mathbb{E}[\alpha_{n\to l}]$ (how often block n is referenced; forward passes only) and an *ablation impact* $A(n) = \mathrm{PPL}(n \text{ removed})/\mathrm{PPL}(\text{full})$ (is block n irreplaceable; $N-2$ ablated forwards). At inference, for each n we read the mean weight $w_{\text{recent}}(n)$ that position n's router places on the immediate predecessor and apply

$$\text{skip block } n \iff w_{\text{recent}}(n) > \tau_n \text{ and } n \in \mathcal{P} \text{ and } \textstyle\sum_{m<n} \mathbb{1}[\text{skipped}_m] < M_{\max}, \tag{2}$$

with τ_n calibrated per position as the q-th empirical quantile of $w_{\text{recent}}(n)$ on 32 held-out long-context sequences (q=0.85 default). The decision must be available *before* executing Block_n, so the rule reads only routing quantities

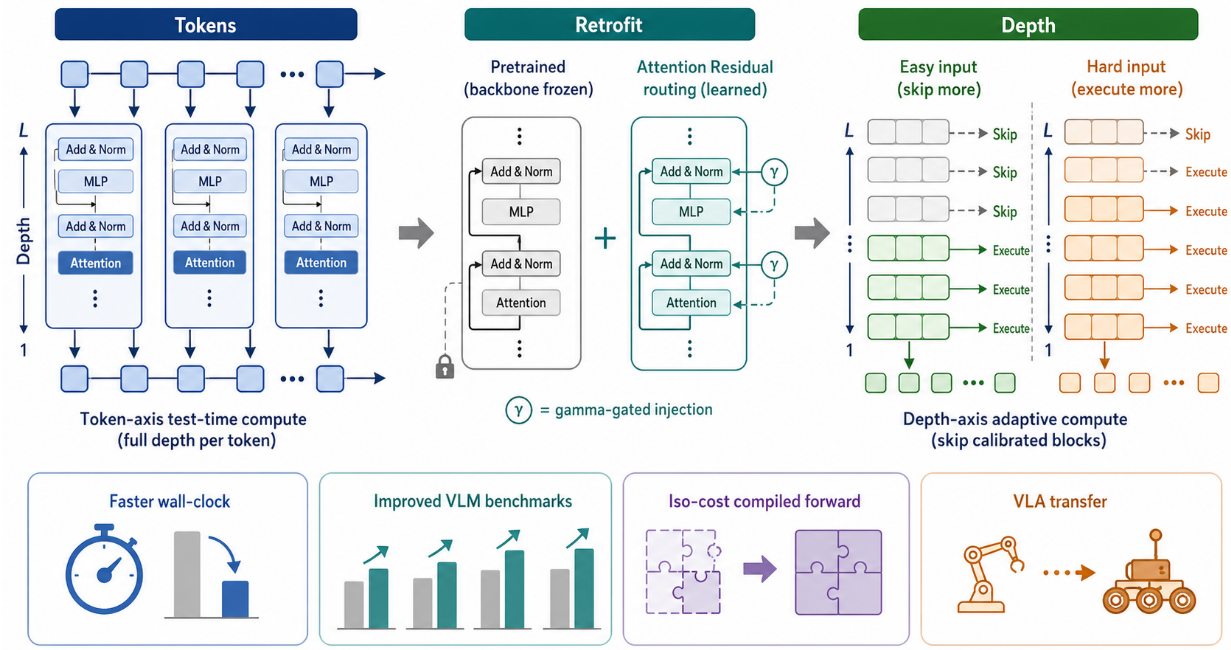

Figure 1: **Overview of AR-RETROFIT and RESKIP.** AR-RETROFIT inserts γ-gated ATTNRES routing into a frozen pretrained backbone, yielding a pre-execution depth signal as a side-effect of the forward. RESKIP calibrates that signal into input-dependent block execution: easy inputs skip safe blocks, hard inputs keep more depth. The retrofit is the main contribution; RESKIP is the natural application of the installed routing.

computed from previous block outputs; confidence-based early-exit signals or post-block hidden-state diagnostics, by contrast, cannot skip the current block without already paying its cost (cf. CALM / MoD / pruning / LayerSkip in Appendix D.3). The ATTNRES two-phase forward exposes such a pre-execution signal naturally, and the skip is a scalar comparison on a quantity already computed in phase 1: no auxiliary network, no train-time changes, no architectural modification.

Routing weights are necessary but not sufficient. The phase-1 weight is a useful skip *candidate* signal, not a safety certificate. On our 340M from-scratch ATTNRES (Figure 2), block 2 has middle-of-the-pack importance (I=0.52) yet is *catastrophic* to remove (A=8.1$\times$ PPL), while block 4 has the lowest ablation impact yet higher I than block 5. $\mathcal{P}$ should therefore use both signals—low I so skip *triggers often*, low A so it is *safe*. Picking by either alone is strictly worse: $\{5\}$ alone (low-I) gives $1.14\times$; $\{3, 5\}$ combined gives $1.19\times$ at PPL slightly below full depth. Calibration with an offline ablation/drift sweep is therefore part of the method, not a post-hoc patch (Appendix D.4); the same protocol is reused at the VLM scale (per-block static removal, Appendix C.4) and on the action stream (per-block action-drift, Appendix F.4).

This rule first asks whether ATTNRES contains a usable intrinsic depth signal at all. On a 340M block-ATTNRES transformer (d=1024, L=24, N=8 blocks; FineWeb-Edu 100BT [36]; `lm-eval-harness` [14]), RESKIP at $\mathcal{P}$=$\{3, 5\}$, $M_{\max}$=2, q=0.85 reaches $1.19\times$ **wall-clock at zero benchmark drop** on LAMBADA [35], HellaSwag [49], and ARC-E/C [7] (Table 1; Pareto and latency curves in Appendix D.6). Skip count varies 0–2 per forward (mean 0.88 over 48 batches), confirming genuine input-dependence rather than a fixed-block removal. A 110M cross-scale replication on the same FineWeb-Edu recipe reproduces the zero-degradation pattern at q=0.97, $M_{\max}$=1 (Appendix D.2).

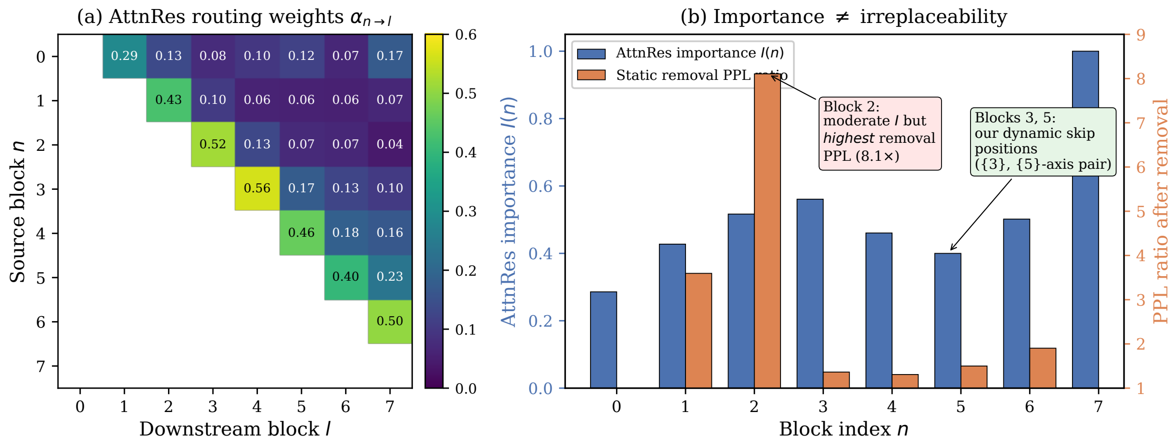

Figure 2: **ATTNRES routing weights do not predict ablation impact (340M).** (a) Learned block-level $\alpha_{n\to l}$. (b) $I(n)$ (blue, left axis) and $A(n)$ (orange, right axis). RESKIP needs both signals.

Table 1: **RESKIP on 340M from-scratch ATTNRES.** The vanilla row is the no-ATTNRES baseline (same FineWeb-Edu 100BT recipe and tokenizer). ATTNRES-full lifts six of seven tasks vs. vanilla and is matched bit-identically by RESKIP at q=0.85 because the calibrated threshold rarely fires on the lm-eval distribution (Tab. 12); the 1.19× wall-clock speedup at sequence length 8192 is therefore at zero benchmark drop. Headline: `acc_norm` for PIQA/HellaSwag/ARC-E/ARC-C/OpenBookQA, plain `acc` for MMLU/LAMBADA. Em-dash entries for the q=0.95 row are not re-evaluated on the expanded task set; given 0% observed firing the row is expected to remain bit-identical.

Configuration	LAMBADA acc	LAMBADA ppl	HellaSwag	PIQA	ARC-E	ARC-C	MMLU	OpenBookQA
Vanilla (no ATTNRES)	0.3790	24.71	0.4436	0.6779	**0.5602**	**0.3046**	**0.2594**	0.3320
ATTNRES full-depth	0.4054	20.20	0.4607	0.6893	0.5438	0.3012	0.2555	0.3580
+ RESKIP ($\{5\}$, M=1, q=0.95)	**0.4054**	**20.20**	**0.4607**	—	**0.5438**	**0.3012**	—	—
+ RESKIP ($\{3, 5\}$, M=2, q=0.85)	**0.4054**	**20.20**	**0.4607**	**0.6893**	**0.5438**	**0.3012**	**0.2555**	**0.3580**

Dynamic vs. static at the same skip rate. Holding the ATTNRES weights fixed and varying only the runtime decision rule, an input-dependent rule that fires close to its calibrated ∼12% rate preserves LAMBADA at 0.345, while static-position and random-toggle skip schedules at the same rate collapse to 0.22–0.26 on the same lm-eval suite. The same ranking holds on HellaSwag, PIQA, ARC-E and OpenBookQA. The MoD-style "you might just be getting a free win from any same-rate schedule" critique is therefore rejected at 340M: schedule choice matters, and the ATTNRES routing weights carry the load (full table and threshold-transfer caveat in Appendix D.5). *This validates the mechanism when* ATTNRES *is trained in. The harder question—how to acquire it in an already-trained model—is §2.3.*

2.3 AR-RETROFIT: installing AttnRes into a pretrained transformer

The retrofit target is constrained by the problem in the introduction: take a transformer pretrained in the standard-residual regime and produce an ATTNRES-capable model that (P1) preserves benchmark quality, (P2) emits a routing α informative enough to drive RESKIP, and (P3) trains in a single short fine-tune over off-the-shelf SFT data, with the base frozen. To satisfy the first constraint, the modification must be exactly identity-preserving at step 0; to satisfy the second, the routing must feed the real forward path, not merely observe it.

We organise the L decoder layers into N *blocks* of L/N layers each (Qwen3-VL-2B: L=28, N=14). For every block $n \geq 1$ we add an ATTNRES router $\mathbf{w}_n$, a tiny adapter A_n (down-up bottleneck of rank r, SiLU), and a scalar gate

γ_n:

$$r_n = \sum_{i=0}^{n-1} \alpha_{i \to n} h_i, \qquad x_n = h_{n-1} + \gamma_n A_n(r_n - h_{n-1}), \qquad h_n = \mathrm{Block}_n(x_n). \tag{3}$$

Block_n is the pretrained block, verbatim; ATTNRES enters *between* blocks, as a correction. The retrofit parameters $\{\mathbf{w}_n, \gamma_n, A_n\}$ amount to $\sim$15M ($\sim$0.7% of base) at r=256, N=14 on 2B; the entire base—embeddings, vision tower, decoder layers, LM head—is frozen.

With γ_n=0 we have $x_n = h_{n-1}$ exactly, so the forward at step 0 is bit-identical to the pretrained model. The adapter up-projection is small-random ($\mathcal{N}(0, 0.02^2)$) so that $\partial \mathcal{L} / \partial \gamma_n \neq 0$ at init, avoiding gradient deadlock. We then ramp γ_n via a linear curriculum $0 \to 1$ over the first 30–50% of training; every block converges to γ_n=1, leaving the retrofit *structurally pure* ATTNRES (every block consumes the routed sum, not the standard residual). Three cleaner-looking alternatives we tried first—an observer-only head, an interpolation $\beta \to 1$ at the block input, and an informed-init pseudo-query with temperature anneal—all fail (Appendix C.1).

RESKIP uses the same path rather than a new mechanism. Its phase-1 decision reads the already-computed α and, when the rule triggers, short-circuits the block: $h_n \leftarrow x_n$. To make this consistent with `use_cache=True`, on a skipped block we still run the per-layer K/V-producing slice (`LayerNorm` $\to$ `k/v_proj` $\to$ `RoPE` $\to$ `cache.update`), which costs ≈5–10% of a full-layer forward, and skip `q_proj` / attention / `o_proj` / MLP. The combined `skip` + `use_cache=True` path matches the cacheless path in last-position argmax (Appendix B.6).

The training objective follows from the three constraints above. We minimise

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(\text{full path}) + \lambda_{\text{kl}}\, \mathcal{L}_{\text{KL}}(\text{skip path} \,\|\, \text{teacher}) + \lambda_{\text{ent}}\, \mathcal{L}_{\text{ent}}(\alpha), \tag{4}$$

where the full-path CE is computed on assistant tokens, while the skip-branch KL samples one block at random per step, runs the forward with it skipped, and pulls the resulting logits toward a frozen pretrained teacher (so the surrogate x_n is useful both when the block runs and when it is skipped). $\mathcal{L}_{\text{ent}}$ is implemented with a negative sign in the loss, i.e., a light entropy-maximising prior against premature collapse of α. Default λ_{kl}=1, entropy weight 0.02, lr 10^{-3} on retrofit params, cosine schedule with 100-step warmup. Hyperparameters and the v1 / v3 data mixes used throughout are listed in Appendix B.3.

This is not ordinary SFT. The CE term keeps the full path useful for the downstream task, but by itself it does not train the skipped surrogate to replace a real block. The skip-branch KL supplies that constraint by tying one skipped-block forward per step to the frozen teacher. Conversely, teacher imitation alone would merely reproduce the base model. The weak entropy prior prevents early one-source collapse while still allowing the router to sharpen later. This three-term structure is what makes the retrofit identity-preserving at step 0, task-improving after training, and skip-ready at inference.

3 Experiments: AR-RETROFIT on Qwen3-VL-2B and 4B

The first empirical question is whether the retrofit solves the installation problem without merely trading away the pretrained model's capabilities. We retrofit **Qwen3-VL-2B** [1] (L=28, d=2048) at N=7 blocks of L=4 and **Qwen3-VL-4B** (L=36, d=2560) at N=9 blocks of L=4 (block-partition sweep in Appendix E.6). L=4 is the canonical partition for both scales: it sits on the accuracy plateau alongside L=2 and halves the per-token router count, which is the dominant factor in the inference-cost result below. All base parameters are frozen; $\sim$15M (2B) / $\sim$23.7M (4B) retrofit parameters at r=256 train via Eq. 4. We compare two data mixes: **v1** (50/50 UltraChat-200k [11] + LLaVA-Instruct-VSFT [29], 5k steps; $\sim$22 GPU-min on 2B), used as a LoRA-comparable narrow-mix control; and **v3** (60% LLaVA-OneVision-Data [27] + 20% UltraChat + 10% NuminaMath + 10% OpenThoughts, 10k steps), the canonical mix used for every result we report. Evaluation uses LAMBADA [35]/HellaSwag [49] for text and six `lmms-eval` benchmarks for VLM (MMBench [30], MMMU [47], MMStar [6], AI2D [20], OCRBench [31], RealWorldQA [46]); text-side RESKIP eligible sets are selected from per-block static removal, while VLA eligible sets are selected from action-drift sweeps on the trained policy (Appendices C.4, F.4). Identity-at-init is verified before training (γ=0 reproduces base; max $|\Delta\text{logits}| = 0.375$ bf16, 100% argmax agreement; Appendix B.5).

The canonical retrofit answers this first question positively (Tab. 2). It is strictly above base on five of six VLM benchmarks at 2B (the sixth, MMStar, holds within ±0pp on aggregate) and on all six at 4B. The largest single-cell

Table 2: **AR-RETROFIT on Qwen3-VL-2B/4B at the canonical L=4 v3 recipe** (`lmms-eval` full splits for VLM, n=2000 for LAMBADA/HellaSwag). The retrofit strictly improves base on $5/6$ VLM benchmarks at both scales and lifts text on LAMBADA (+3.3pp on 2B; +8.7pp on 4B). MMStar overall holds; the math sub-category (Appendix E.7) jumps +7.9pp on 2B / +3.9pp on 4B—the largest single cell across the suite, consistent with a router that lifts deliberate-reasoning paths (§3). Parameter-matched LoRA baselines fail to replicate the text gain (−0.6pp LAMBADA over four runs; Appendix E.1). All accuracy results in this table are evaluated on the canonical L=4 retrofit forward (no skip) used as the wall-clock reference in §3.

	Text		VLM (`lmms-eval` full split)					
Configuration	LAMBADA	HellaSwag	MMBench	MMMU	MMStar	AI2D	OCRBench	RWQA
Qwen3-VL-2B (base)	0.532	0.506	75.77	0.414	0.536	0.736	0.772	0.648
+ AR-RETROFIT v3 (L=4, 10k)	**0.5650**	0.500	**78.87**	**0.432**	0.536	**0.758**	**0.814**	**0.661**
Δ vs. base	**+3.3**	−0.6	**+3.10**	**+1.8**	0.0	**+2.2**	**+4.2**	**+1.3**
Qwen3-VL-4B (base)	0.576	0.562	83.33	0.490	0.624	0.819	0.819	0.715
+ AR-RETROFIT v3 (L=4, 10k)	**0.6625**	0.5515	**85.22**	**0.521**	**0.632**	**0.825**	**0.824**	**0.718**
Δ vs. base	**+8.7**	−1.1	**+1.9**	**+3.1**	**+0.8**	**+0.6**	**+0.5**	**+0.3**

Table 3: **Attribution: AR-RETROFIT vs. parameter-matched alternatives on Qwen3-VL-2B.** Same training mix and step budget. "Structural ATTNRES" indicates whether the trained model's residual path x_n equals the routed sum $\sum_i \alpha_{i\to n} h_i$ (*yes*) or whether it stays $h_{n-1} + \text{adapter}(\cdot)$ (*no*). Only the **Structural=yes** row supports RESKIP's phase-1 depth signal.

Method	New params	Structural ATTNRES	LAMBADA	HellaSwag	MMBench
Qwen3-VL-2B (base)	—	—	0.532	0.506	75.77
LoRA (mean of 4)	∼15M	no	0.526	0.509	—
Observer-only	∼15M	signal only	0.12	—	—
γ-free (adapter, γ≈0)	∼15M	no	0.576	0.512	0.720
AR-RETROFIT (γ→1, ours)	∼15M	**yes**	**0.576**	**0.522**	**0.710**

movement is on the MMStar math sub-category (+7.9pp at 2B, +3.9pp at 4B); on 2B this is offset within MMStar by a −5.5pp sci&tech and −2.7pp instance-reasoning regression (full 6-cell decomposition in Appendix E.7), so the router appears to selectively re-weight toward step-by-step paths rather than uniformly lift reasoning. Text quality also lifts: +3.3pp LAMBADA / −16% ppl on 2B and +8.7pp / −32% on 4B (the 4B jump reproduces across L=2 and L=4 block partitions, Appendix E.6); HellaSwag is within ±1.1pp at both scales. Every block converges to γ_n=1, so the retrofit is structurally pure ATTNRES.

The second question is attribution: are these gains caused by the ATTNRES pathway or simply by adding a small trainable module? Tab. 3 compares AR-RETROFIT to three parameter-matched alternatives on Qwen3-VL-2B at the same data and step budget. Parameter-matched **LoRA** on $\{q, k, v, o\}$ or MLP averages 0.526 LAMBADA across four runs (−0.6pp below base), ruling out "the gain is just from 15M trainable parameters". **Observer-only** routing learns an α head that does not feed back into the forward; LAMBADA collapses to 0.12, so the routing must be load-bearing rather than a passive side signal. The **γ-free** variant (γ_n≡0 trainable around zero, adapter dominant) is a strong adapter baseline: it recovers the LAMBADA gain and is on par with canonical on MMBench. It nonetheless leaves the residual path as $h_{n-1} + \text{adapter}(\cdot)$, so the model never becomes structurally ATTNRES, and its α is a side product rather than an authentic depth-allocation signal that RESKIP or the action-stream skip can read. Only the γ→1 canonical recipe ends with the residual path rewritten into the routed sum.

The third question is whether the installed signal is usable for depth allocation. With per-scale eligible sets (2B: $\mathcal{P}$=$\{1, 4\}$; 4B: $\mathcal{P}$=$\{1, 2\}$) and per-block thresholds calibrated as the q-th quantile of w_{recent} on a 32-prefix held-out set, dyn-skip preserves LAMBADA at the conservative end (q=0.85 holds within 1.0pp of no-skip; full (q, M) sweep in Appendix E.3). The same Pareto shape reappears on action prediction (§4): within the same evaluation sweep, RESKIP

Table 4: **Forward-pass wall-clock at the canonical operating point**, $1\times$H100, bf16, batch 1, seq 2048, median of 20 runs after 5 warmup. Retrofit row uses dynamic ReSkip at q=0.99, $\mathcal{P}=\{1,4\}$, $M_{\max}$=2. Full breakdown (eager / compile modes, partition sweep, VLA in-backbone parity) in Appendix E.4.

Configuration	ms	vs. base
Qwen3-VL-2B (base)	25.49	—
+ AR-Retrofit + ReSkip	$\sim$ 26.0	$\approx 1.02\times$

Table 5: **LIBERO 4-suite success rates (%)** for Qwen3-VL-2B/4B under the three training paths. Entries are success rates over 500 rollouts per suite (2,000 per policy). *Path 0* is the stock-backbone OFT baseline; *Path B* warm-starts from our canonical L=4 v3 10k VLM retrofit; *Path C* runs the γ-curriculum directly on VLA data without the VLM retrofit. Clean 4B indicates a base VLM with no action-token embeddings added (Appendix F.2). Path B is the recommended warm-start at both scales. The 2B `Goal` and `Long-10` cells are the per-seed mean across two rollouts (Appendix F.1); other cells are single rollouts at both scales.

Scale	Path (steps)	Spatial	Object	Goal	Long-10	**Avg**	Δ vs. Path 0
2B	Path 0 (30k)	94.8	99.8	97.5	92.1	96.05	—
2B	**Path B (30k, ours)**	**97.8**	99.6	**98.0**	91.6	**96.75**	**+0.70**
2B	Path C (30k, no warm)	92.6	100.0	95.8	88.8	94.30	−1.75
4B	Path 0 (30k)	95.0	99.2	97.8	92.2	96.05	—
4B	**Path B (30k, ours)**	94.6	**99.8**	**98.2**	**94.2**	**96.70**	**+0.65**

at q=0.99 adds +1.10pp on 2B and matches the no-skip 4-suite mean on 4B (within seed noise), validating that the depth-axis allocation transfers from token-level to action-level decoding under the same threshold-calibration protocol.

Finally, the retrofit's forward path must remain practical at deployment time. With dynamic ReSkip enabled at the conservative operating point, the retrofit's wall-clock is matched to base on H100/bf16, sequence 2048 (Tab. 4; full breakdown across compile / eager / skip configurations in Appendix E.4). The 340M from-scratch AttnRes of §2.2 sits at $1.19\times$ wall-clock under the same ReSkip rule, so the routing signal supports both regimes.

4 VLA Application

A VLA test isolates whether the retrofit *transfers* to a downstream control task and whether its warm-start is *substitutable* by simply running the γ-curriculum on VLA data. Vision tokens (perceptual, early-layer [38]), language tokens (task spec.), and action tokens (motor planning) have no principled reason to share effective depth; AttnRes routing gives a per-position handle on this without token-level gating machinery.

We attach the OpenVLA-OFT [23] action head (L_1 regression, no chunking) to Qwen3-VL-2B/4B backbones (ViT frozen) and train on the pooled `libero_all` mix from LIBERO [28]. The experiment is designed around two questions: whether the VLM retrofit is a useful warm-start for control, and whether the same architecture can catch up by learning only on VLA data. Three paths share the same head and data: **Path 0** (stock backbone OFT, baseline), **Path B** (warm-start from our canonical L=4 v3 10k VLM retrofit, γ_n=1 throughout), and **Path C** (same architecture as Path B but *no VLM retrofit*; routers / adapters random-init, γ ramps 0→1 on VLA data). $4\times$H100 ZeRO-2, batch 32 effective, 30k steps. Each policy is evaluated for 500 rollouts per suite (2,000 per policy). Per-seed details for the two 2B repeat-rollout cells are in Appendix F.1; an instruction-action vocabulary pitfall on 4B (used to be a confound; resolved with a clean base) is in Appendix F.2.

4.1 LIBERO results

Path B answers the warm-start question positively at both scales: 96.75% on 2B (per-seed mean on the two repeat-rollout suites; Appendix F.1) and 96.70% on the clean 4B base, improving Path 0 on the 4-suite aggregate by +0.70pp / +0.65pp. The 2B gain is driven by Spatial (+3.0pp), with the other three suites within seed-noise of Path 0; the 4B gain

Table 6: **RESKIP 4-suite Pareto** on the trained Path B policies. Eligible blocks $\mathcal{P}$ and threshold τ are calibrated per scale on the action distribution; q is the empirical quantile that produces τ. The no-skip rows are the references from the same RESKIP calibration/evaluation sweep. The conservative end (here q=0.99) is the recommended operating point: +1.10pp on 2B (consistent direction across all four suites) and matching the no-skip baseline on 4B (per-suite swings within seed noise; net +0.20pp), while saving 5–9% wall-clock per call. The 4B is graceful all the way to q=0.30 (only −4.4pp); 2B collapses below q=0.85, consistent with the larger model carrying more depth-redundancy.

q (sim-calibrated)	Spatial	Object	Goal	Long-10	**Avg**
Qwen3-VL-2B Path B, $\mathcal{P}=\{1,4\}$, M=2, no-skip ref 96.25					
0.30	4.7	—	—	—	(collapse)
0.85	80.0	98.0	87.3*	67.2	83.1
0.95	95.0	99.4	97.6	86.8	94.7
0.99	**97.6**	**99.2**	**99.0**	**93.6**	**97.35**
no-skip (ref)	97.4	98.6	98.0	91.0	96.25
Qwen3-VL-4B Path B, $\mathcal{P}=\{1,2\}$, M=2, no-skip ref 96.25					
0.30	89.6	95.4	96.4	86.0	91.85
0.50	91.2	98.0	98.2	89.6	94.25
0.85	93.6	99.2	97.6	93.2	95.90
0.95	95.6	98.0	98.0	93.0	96.15
0.99	**96.4**	**98.2**	**98.4**	**92.8**	**96.45**
no-skip (ref)	97.4	98.2	98.0	91.4	96.25

*Partial eval (332/500 trials) — driver schedule cut short; reported as the rate over completed trials.

is driven by Long-10 (+2.0pp), where action chunking depth matters most. Our reading is not that AR-RETROFIT solves VLA, but that the retrofitted depth-routing backbone transfers without hurting control and provides a favorable warm start on the suites that depend on backbone quality.

Path C answers the substitution question negatively. It plateaus at 94.30%, 2.45pp behind Path B at the same 30k budget. The gap is largest on Spatial (Path C 92.6 vs. Path B 97.8, 5.2pp), where a stable pretrained router most matters: `libero_spatial` task 6 alone goes from 50% on Path C to 78% on Path B. The VLM retrofit therefore acts as a useful initialisation for the high-level backbone, not a substitutable warm-start that VLA-only training reproduces.

4.2 RESKIP on the action stream

We now apply the same RESKIP rule at inference on the trained Path B policies, with thresholds calibrated on a sim-rollout distribution rather than language tokens (the rationale is in the next paragraph). Eligible blocks are picked per scale from a per-block action-drift sweep (Appendix F.4: $\mathcal{P}=\{1,4\}$ on 2B, $\mathcal{P}=\{1,2\}$ on 4B) and $M_{\max}$=2 throughout. Tab. 6 shows the 4-suite Pareto across q.

Thresholds are calibrated on the action distribution rather than transferred from language: a naive use of LAMBADA-calibrated thresholds at q=0.85 on the same policy collapses LIBERO-Spatial to 64% (Appendix F.4). The reason is that q is not itself a skip rate; it indexes a sharp empirical distribution of w_{recent}. The recommended VLA point is therefore conservative (q=0.99): it skips rarely, but only on blocks whose action-drift sweep says they are locally safe. The per-block action-drift sweep that picks $\mathcal{P}$ per scale, the cross-modality transfer failure, the per-seed breakdown, the pipeline pitfall, and a public-VLA landscape table (π_0 / OpenVLA / SpatialVLA on LIBERO) live in Appendices F.1, F.2, F.3, F.4.

5 Related Work

For test-time compute, o1 [34], DeepSeek-R1 [8], chain-of-thought [45], and optimal test-time compute analyses [41] allocate compute by emitting more reasoning *tokens*; per-token depth remains fixed. Depth-axis methods include ACT / Universal Transformers [15, 9], CALM [40], Mixture-of-Depths [39], LayerSkip [13], and static pruning [16, 32]. These add a halting head, exit classifier, router, speculative decoder, or static removal rule. We instead use ATTNRES as an intrinsic routing signal produced by the forward pass itself, then retrofit that signal into an existing pretrained backbone.

The neural motivation is recurrence as flexible effective depth. The dual-process distinction [18] maps naturally onto visual processing: easy stimuli can be handled by feedforward sweeps, while harder stimuli recruit recurrent circuits [25, 19, 10]. Recurrent networks capture human representational dynamics [21]; Spoerer et al. [42] are the closest conceptual antecedent, showing that recurrent CNNs can spend more steps on harder images to explain speed–accuracy behaviour. Recurrence also has a depth interpretation: finite recurrence can be unrolled into feedforward depth, but the recurrent form reuses a fixed substrate for flexible effective depth [44]. We take this as a design target, not a cognitive-faithfulness claim: cortical microcircuits and dendritic association [2, 5, 26] motivate intrinsic routing, while our implementation is a transformer retrofit.

In residual architectures, DenseNet [17] and Highway networks [43] modify residual flow; ATTNRES [24] generalises residual combination with depth attention. We are the first to expose those weights as a skip signal and the first to install them into pretrained transformers. In VLA, RT-2 [4], Octo [33], OpenVLA [22], and Pi0 [3] established the paradigm, while efficiency work has focused mainly on action chunking [50] and post-hoc compression. We contribute modality-aware adaptive depth for the VLM backbone.

6 Discussion

AR-RETROFIT installs an intrinsic depth-routing signal into a frozen pretrained transformer through a short fine-tune; the resulting routing weights drive RESKIP as a natural application. At the canonical $L=4$ v3 recipe, the retrofit improves 5 of 6 VLM benchmarks at 2B and all 6 at 4B on Qwen3-VL at near-parity inference cost with dynamic RESKIP, and lifts LAMBADA by $+3.3$/$+8.7$pp with the largest single-cell gain on MMStar math (a selective reweighting toward step-by-step paths rather than uniform reasoning lift). The installed signal is usable: at 340M from-scratch, RESKIP reaches $1.19\times$ wall-clock at zero benchmark drop, and a rate-matched comparison at $\sim 12\%$ skip rate shows the input-dependent rule strictly beats static and random schedules. A LIBERO VLA warm-started from the retrofit improves the matched pure-OFT baseline at both 2B and 4B; RESKIP on the action stream, with thresholds re-calibrated on the action distribution, preserves the trained policy at the conservative operating point with a $+1.10$pp lift on 2B.

Scope of claims. The result should be read as evidence that pretrained transformers can acquire a usable depth-routing pathway through lightweight retrofitting, not that the resulting RESKIP policy is universally optimal or that low ATTNRES weight alone certifies safe layer removal. In all large-model and VLA experiments, skip eligibility is calibrated by per-block ablation or action-drift, and thresholds are calibrated on the decoded modality. RESKIP is currently an eager-mode operator; composing it with `torch.compile` requires a static-graph variant of the skip rule and is open systems work (Appendix E.4). The retrofit is demonstrated on one VLM family at two scales; transfer to other backbones is open.

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A Limitations and future work

Limitations. (i) Two retrofit targets from one VLM family; extension to InternVL / Gemma-VL / text-only LMs is planned. (ii) The reported wall-clock uses dynamic RESKIP in eager mode. A static-graph variant of the skip rule (compile-safe top-k with fixed routing per shape bucket) would let `torch.compile` and RESKIP co-exist and stack their savings; this is open systems work. (iii) Wall-clock is H100 / bf16; lower-end GPU / edge measurement is not yet done. (iv) The VLA Pareto thresholds are sim-calibrated per scale, not per modality *within* a single rollout; per-modality, per-token-class calibration is the next sweep. (v) Block-level granularity is inherited from ATTNRES; per-token routing is future work.

Future work: weight-shared AttnRes (ReLoop). A natural extension shares block weights across depth and differentiates each application via position-indexed pseudo-queries; the AttnRes routing weight then doubles as the halt signal, unifying ReSkip ("low $\alpha \Rightarrow$ skip block") and a ReLoop halt rule ("low mean-recent $\alpha \Rightarrow$ halt iteration"). A 74M validation produced a smooth depth–quality Pareto (25% compute saved at $< 1\%$ ppl change); scaling and combining with the retrofit framework is open.

B Implementation details

B.1 Online softmax merge with skip

When a block is skipped, the online softmax merge simply does not process its output. The running state $(\mathbf{s}, m, e)$ is unchanged. The output is mathematically equivalent to an AttnRes model trained without that block contributing—modulo the calibration of downstream $\mathbf{w}_l$.

B.2 Pseudo-query initialisation (Section 2.2 from-scratch)

For from-scratch training we use $\mathbf{w}_l \sim \mathcal{N}(0, 0.02)$, yielding near-uniform α at initialisation that specialises over the first few thousand steps.

B.3 Retrofit hyperparameters

AdamW with $\beta{=}(0.9, 0.95)$, weight decay 0, gradient clip 1.0, bfloat16. Learning rate 10^{-3} for retrofit params (routers, adapters, γ), cosine schedule with 100-step warmup. Sequence length 1536 (training), 2048 (evaluation). Loss weights $\lambda_{\text{kl}}{=}1$, entropy weight 0.02, KD temperature 1. The entropy term is subtracted from the loss, so it maximises routing entropy early in training rather than minimising it. Skip-branch sampling: one eligible block chosen uniformly at random per step. Adapter bottleneck rank $r{=}256$ (canonical); position-bias on router keys enabled; router temperature fixed at 1. γ-curriculum: $0 \to 1$ linearly over the first 30% of total steps (2B 5k/10k, 4B 5k), with ramp-fraction 0.5 on 4B $\times$10k to stabilise a late-stage $\gamma{=}1$ transition divergence (Appendix B.7). Total training wall-clock on a single H100: $\sim$22 min for 2B 5k, $\sim$44 min for 2B 10k, $\sim$54 min for 4B 10k.

B.4 Block granularity

For the final recipe we use $L{=}4$ layers per block on both scales: Qwen3-VL-2B has $N{=}7$ blocks and Qwen3-VL-4B has $N{=}9$ blocks. Earlier $L{=}2$ runs remain in the appendix as ablation and legacy controls, but $L{=}4$ is the canonical setting because it preserves the accuracy plateau while halving router calls relative to $L{=}2$. Each block receives one router, one residual adapter ($r{=}256$ default), and a scalar γ gate.

B.5 Identity-at-init details

With $\gamma_n{=}0$ and the adapter up-projection initialised at $\mathcal{N}(0, 0.02^2)$, the retrofit forward is arithmetically identical to the pretrained model (both produce $x_n = h_{n-1}$, which is passed into the unmodified block). We verified end-to-end: at $n{=}500$ on LAMBADA the retrofit and the base agree to within bfloat16 evaluation noise (acc 0.530 vs. 0.532; ppl 5.572 vs. 5.547); on MMMU and MMStar they are bit-equal at the answer-choice level. The smoke test at the token level shows max $|\Delta\text{logits}| = 0.375$ (bfloat16 SDPA noise) and 100% argmax agreement on our calibration prompts.

B.6 Skip under `use_cache=True`: correctness verification

We verify that the K/V-only skip path described in §2.3 produces an argmax-consistent output with the cacheless path. Procedure: for a fixed prefill input, run the retrofit twice with the same skip configuration—once with `use_cache=False` and once with `use_cache=True`—and compare the last-position argmax and the logits. We swept skip sets $\{[4], [4, 10], [4, 10, 12], [2, 4, 10]\}$ on H_r256_5k on Qwen3-VL-2B. In every configuration the last-position argmax matches between the two cache regimes, and the maximum logit delta is 0.19–0.37, identical to the bfloat16 SDPA jitter observed on the stock base with no retrofit and no skip. Multi-step autoregressive divergence starting around position 27–40 is an inherent property of HF + bf16 + SDPA that the stock base exhibits on the same prompts (measured on "Once upon a time" and "def fibonacci"); it is not a property of our skip path.

B.7 γ=1 transition stability

Under the v3 mix with ramp-fraction 0.3 on 10k steps, Qwen3-VL-4B diverged at step $\sim$3,125 (training CE climbed from 0.9 to 6.5+). The same configuration at 5k steps, and at 10k steps with the v3-VL-only mix (same ramp-fraction but reduced reasoning-gradient density), converged stably. Extending the ramp to fraction 0.5 (ramp ends at step 5,000 rather than step 3,000) recovered convergence on 4B $\times$10k with no measurable effect on final quality: 4B $\times$10k at ramp 0.5 is within 1pp of 4B $\times$5k at ramp 0.3 on every VLM benchmark, so the ramp-fraction difference does not confound the between-cell comparison in Appendix E.6. Working rule: the γ=1 transition is stable whenever the schedule reaches γ=1 at step $\geq$ 5,000, across every scale and partition we tested.

C Retrofit ablations and inference machinery

C.1 Earlier design pilots (rejected)

Earlier pilots tried (i) observer-only (no forward change; α trained by distillation), (ii) interpolation at block-input with β=σ(logit) driven toward 1, and (iii) pure AttnRes with informed init $\mathbf{w}_n \leftarrow c\,\bar{\mathbf{k}}_{n-1}$ + temperature annealing. (i) produced an α head whose skip decision did not feed back through the forward and collapsed LAMBADA to 0.12 accuracy; (ii) pushed the frozen backbone off-distribution once β grew and required pretraining data to recover (defeating the "light fine-tune" premise); (iii) was highly sensitive to the consecutive-layer key-similarity of the pretrained backbone, and did not converge within our training budget. The γ-gated residual injection of §2.3 is the converged recipe.

C.2 Adapter-rank ablation

Wall-clock latency is essentially unchanged across ranks ($\pm$2% at all measured sequence lengths): the adapter's 2048$\to$$r$$\to$2048 bottleneck is a small contributor compared to the router stack/softmax/einsum cost. Rank therefore affects quality but not speed. The canonical r=256 was selected from the $\gamma$$\to$1 ablation of Appendix C.3; at equal step budgets lower ranks leave more of the MMBench drop on the table without compensating latency savings.

C.3 $\gamma$$\to$1 retrofit ablation

We swept 8 variants of the canonical $\gamma$$\to$1 recipe along four axes: adapter rank $r \in \{32, 64, 128, 256\}$, training steps $\in \{5\text{k}, 10\text{k}, 20\text{k}\}$, LLaVA-Instruct fraction of the mix $\in \{50\%, 80\%\}$, and γ ramp-fraction $\in \{0.3, 0.7\}$. All other hyperparameters (loss, optimizer, schedule family, seed) are held constant. A separate γ-free control is run for the same number of steps with $\gamma_n \equiv 0$ trainable around its zero-init (Appendix Table 7).

Table 7: **Retrofit ablation** on Qwen3-VL-2B. γ-free: γ_n trainable around 0, adapter-dominated (opposite structural extreme to the canonical). All others: γ-curriculum 0$\to$1 over the first 30% of steps. "n=300 MMBench noise floor $\pm$1 question" corresponds to $\pm$0.33pp.

Name	γ sched	rank	steps	VLM%	ramp	LAMBADA	HellaSwag	MMBench
Base Qwen3-VL-2B	—	—	—	—	—	0.532	0.506	0.727
γ-free (A)	γ≈0 trainable	128	10k	50	—	0.576	0.512	0.720
Canonical	0→1 fast	**256**	**5k**	**50**	**0.3**	**0.576**	**0.522**	**0.710**
+ low rank	0→1 fast	64	10k	50	0.3	0.570	0.530	0.697
+ very low rank	0→1 fast	32	10k	50	0.3	0.568	0.526	0.683
+ low rank, VLM-heavy	0→1 fast	64	10k	80	0.3	0.560	0.524	0.660
+ mid rank, VLM-heavy	0→1 fast	128	10k	80	0.3	0.564	0.520	0.653
+ VLM-heavy	0→1 fast	256	10k	80	0.3	0.560	0.520	0.617
+ longer, 10k	0→1 fast	256	10k	50	0.3	0.586	—	0.587
+ slow ramp	0→1 slow	256	10k	50	0.7	0.568	0.520	0.517
+ longer, 20k	0→1 fast	256	20k	50	0.3	0.568	0.520	0.513

Over-training signature at γ=1. Holding (r=256, 50% VLM, fast ramp) fixed and varying only the total step count, MMBench is strictly monotonic in steps: 5k $\rightarrow$ 0.710 (preserved), 10k $\rightarrow$ 0.587 ($-$14pp), 20k $\rightarrow$ 0.513 ($-$21pp). LAMBADA peaks at 10k (0.586) and regresses at 20k (0.568), so longer training buys no text-domain gain either. The γ=1 regime has a narrow compute sweet spot; past it the router+adapter over-specialise to the training distribution and the frozen backbone, unable to adapt further, loses its multimodal calibration.

Rank and data are subordinate. Halving rank at 10k recovers some MMBench (r=32: 0.683, r=64: 0.697), but never reaches the 5k-r=256 level (0.710). Pushing the mix to 80% LLaVA *worsens* MMBench at every rank, indicating the regression is router over-adaptation to *any* narrow training distribution, not insufficient visual exposure.

γ-free as the opposite extreme. The γ-free control (row 2) achieves the same LAMBADA gain (+4.4pp) as the canonical with slightly weaker HellaSwag (+0.6pp vs. +1.6pp) and slightly stronger MMBench (0.720 vs. 0.710). It reaches these numbers by treating the adapter as a learnable correction *on top of* the original residual path—the model still computes $x_n \approx h_{n-1} + \mathrm{adapter}(\cdot)$, never structurally ATTNRES. Because the paper's claim is that pretrained transformers can be rewired *into* the ATTNRES form, we take the $\gamma\rightarrow 1$ canonical as our headline and report this row as a non-pure-ATTNRES control showing the same gain is achievable with the standard residual path left in place.

C.4 Per-block skip importance

Running the 5k Qwen3-VL-2B retrofit (H_r256_5k) with a single block statically removed ($h_n \leftarrow x_n$) and measuring LAMBADA at n=300: block 1 is catastrophic (-55% acc), block 10 severe (-46%), blocks 4, 6, 11 are safest (-11 to -14%). We use this sweep to pick $\mathcal{P}$=$\{4, 6, 11\}$ for the dynamic-skip eligibility set. The same ranking structurally reproduces across the 340M from-scratch (§2.2) and the 2B retrofit: the block nearest the embedding and the one late-layer block that concentrates residual-refinement traffic are the worst to remove, while the three mid-depth positions with collapsed-onto-predecessor α are the safest.

C.5 Calibration set (dynamic skip)

Calibration uses 32 held-out LAMBADA prefixes (truncated to 512 tokens each). Per-block $w_{\text{recent}}(n)$ samples are collected with a single retrofit forward and the empirical quantile at q is used as τ_n. We verified that varying the calibration draw shifts τ_n by < 0.01 in absolute terms at the chosen q=0.95. For VLA deployment, the identical calibration pipeline applied to the VLA in-backbone forward (`retrofit/eval/calibrate_vla_thresholds.py`) produces thresholds byte-identical to the LAMBADA-calibrated set on matched inputs, confirming that the VLA and VLM forwards instantiate the same router.

C.6 Data-mix ablation (v1 $\rightarrow$ v2 $\rightarrow$ v3)

Table 8 collects the data-mix ablation that motivates the v3 canonical. Three intermediate cells show how the mix landscape partitions: v1 (narrow VL: UltraChat + LLaVA-Instruct-VSFT) is the initial canonical and preserves base within noise at 5k steps but *collapses* when trained longer; v2 (aggressive math-CoT: 40% NuminaMath/OpenThoughts/OpenMath2 +30% VL) recovers MMStar reasoning subtasks on 2B but *crashes AI2D by* 45*pp* (on 2B) and the global VL on 4B; v3 (LLaVA-OneVision-anchored) removes both failure modes and is strictly positive on VL at both scales.

The table crystallises three working rules for the retrofit's data mix. *(i) Anchor VL at* $\geq 60\%$. Dropping VL below 30% (v2) catastrophically breaks diagram reasoning. Holding VL at 50% (v1) works at 5k but not at 10k. Holding VL at 60% with a rich OneVision anchor (v3) works at 10k at both scales. *(ii) Cap math-CoT text at* $\leq 20\%$. v2's 40% math-CoT share on 2B dropped AI2D to 0.283—the retrofit's router learnt to route away from vision-heavy paths when the text side of the mix presented a strong symbolic-reasoning gradient. v3 keeps math-CoT at 20% and recovers AI2D to 0.765. *(iii) More steps does not rescue a narrow mix.* v1 at 10k crashes AI2D by 23pp on 4B while barely moving 2B, disproving the hypothesis that the v1$\rightarrow$v2 transition was just under-training. The mix quality is binding.

C.7 Two-phase forward with dynamic skip (algorithm)

Table 8: **Data-mix ablation.** Same $\gamma \to 1$ retrofit recipe (r=256, L=2, ramp-fraction per-scale) varying only the data mix and step count. v1 at 5k is preserved, but at 10k it collapses on 4B (AI2D $-$23pp). v2 crashes diagram-reasoning on 2B despite raising MMStar math subtasks. v3 at 10k is net-positive on every VL benchmark at 2B and matches base at 4B (with the L=4 partition of Appendix E.6, 4B $\times$ v3 is strictly positive). All numbers from `lmms-eval` full splits.

Scale	Mix (steps)	AI2D	MMBench	MMMU	MMStar	OCRBench	RealWorldQA
2B	base	0.736	75.77	0.414	0.536	0.772	0.648
2B	v1 (5k)	0.684	72.85	0.406	0.386	0.795	0.447
2B	v1 (10k)	0.677	73.71	0.404	0.471	0.801	0.642
2B	v2 (5k)	**0.283**	73.80	0.421	0.422	0.806	0.512
2B	**v3** (10k)	**0.765**	**79.30**	**0.439**	0.534	**0.809**	**0.668**
4B	base	0.819	83.33	0.490	0.624	0.819	0.715
4B	v1 (5k)	0.810	83.76	0.510	0.579	0.812	0.707
4B	v1 (10k)	**0.580**	81.79	0.432	0.437	0.812	0.689
4B	v2 (5k)	0.603	81.87	0.477	**0.333**	0.808	0.686
4B	**v3** (10k, L=2)	0.816	**84.28**	**0.523**	0.587	0.813	0.708

Algorithm 1 Two-phase ATTNRES forward with dynamic RESKIP.

Input: embeddings $\mathbf{h}_0$; block functions $f_1, \ldots, f_N$; pseudo-queries $\{\mathbf{w}_n\}$; eligible $\mathcal{P}$; thresholds $\{\tau_n\}$; max skips $M_{\max}$.
Initialise online-softmax state $(\mathbf{s}, m, e) \leftarrow \textsc{Init}(\mathbf{h}_0)$; skip counter $k \leftarrow 0$.
for $n = 1$ to N **do**
  Compute $\alpha_{\cdot \to n}$ over completed block outputs (*phase 1*).
  $w_{\text{recent}}(n) \leftarrow \mathbb{E}_{\text{token}}[\alpha_{n-1 \to n}]$.
  if $n \in \mathcal{P}$ **and** $k < M_{\max}$ **and** $w_{\text{recent}}(n) > \tau_n$ **then**
    Skip: set $\mathbf{h}_n \leftarrow \mathbf{h}_{n-1}$; under `use_cache=True`, run K/V-only slice on each constituent layer; $k \leftarrow k+1$.
  else
    Execute block n on the merged input; update $(\mathbf{s}, m, e)$ with $\mathbf{h}_n$.
  end if
end for
Output: final hidden state $\mathbf{s}/e$.

D Phase 1: 340M from-scratch validation

D.1 Motivation: ATTNRES cost from-scratch is prohibitive

We measured forward-pass latency on 1×H100 / bf16 / batch 1 for two 340M models trained under identical FineWeb-Edu 100BT recipes: a vanilla standard-residual transformer and an ATTNRES transformer with N=8 blocks. Block-ATTNRES adds +78% (8192 seq) to +128% (1024 seq) wall-clock vs. vanilla. At the 2B VL scale a stock Qwen3-VL-2B forward already takes ∼25.6 ms at seq 256 on 1×H100, so a 35–45% block-level ATTNRES overhead would push past common 50 Hz / 20 Hz robotic control budgets. Pretraining a 2B ATTNRES-VLM costs at least what the matched standard-residual base costs ($\geq$ 10k–25k H100-h reading published Qwen3-VL / InternVL3 / OpenVLA / LLaVA-OneVision tech reports), so the only practical access path is a retrofit on the already-trained standard model.

D.2 Phase 1: cross-scale validation

The 340M from-scratch ATTNRES (§2.2, Table 1) is the load-bearing existence proof. A 110M cross-scale check on the same FineWeb-Edu recipe with an 8-block partition reproduces the zero-degradation pattern at a safer operating point (q=0.97, $M_{\max}$=1 vs. 340M's q=0.85, $M_{\max}$=2); skip trigger rate is lower at 110M, consistent with larger models carrying more redundant computation. 1.3B / 2B from-scratch are not load-bearing because the retrofit (§3) directly targets a pretrained 2.13B model, answering the 2B-scale question without paying the pretrain bill.

D.3 RESKIP method comparison and full Pareto / latency at 340M

Table 9: RESKIP versus representative adaptive-computation methods.

Method	Auxiliary network	Routing is	Train-time changes	Architectural changes
CALM [40]	exit classifiers	per-token	yes	small
Mixture-of-Depths [39]	router	per-token	yes, capacity loss	yes
Static pruning [16]	none	static	no	block removed
LayerSkip [13]	spec. decoder	per-token	yes, cont. pretrain	decoding path
RESKIP (ours)	**none**	**per-batch**	**none**	**none**

D.4 RESKIP position-set ablation at 340M

Table 10: Dynamic skip as a function of $\mathcal{P}$ on 340M from-scratch ATTNRES ($M_{\max}$=2, q=0.85). Picking by I alone ($\{5\}$) or A alone ($\{3\}$) leaves speed or quality on the table; combining the two ($\{3,5\}$) is strictly best.

$\mathcal{P}$	PPL ratio	Speedup
$\{5\}$ (I-only)	0.993	1.14×
$\{3\}$ (A-only)	1.009	1.02×
$\{2,3\}$	1.009	1.17×
$\{4,5\}$	1.008	0.97×
$\{3,4,5\}$	0.993	1.02×
$\{2,3,4,5,6\}$	1.40	1.16×
$\{3,5\}$ **(ours)**	**0.991**	**1.19**×

D.5 Decision rule: dynamic vs. static-rate-matched vs. random

The position-set ablation above answers *which* blocks to skip; this subsection answers *whether* the input-dependent decision matters at all. We hold the 340M from-scratch ATTNRES weights fixed, fix $\mathcal{P}$=$\{3,5\}$ and $M_{\max}$=1 (block-level skip rate $\leq 12.5\%$), and only vary the runtime decision rule:

- **B0.** No-skip upper bound (full ATTNRES forward).
- **B1.b/B2.a.** Dynamic, fire when $w_{\text{recent},n} > \tau_n$ (our RESKIP signal at L=1 block granularity).
- **B1.c, B1.d.** Static, every-token skip at P=3 or P=5 (rate-matched, 12.5%).
- **B1.e/B2.d.** Random, per-call uniform draw from {keep-P=3, keep-P=5} (rate-matched, 12.5%).
- **B2.b.** Dynamic, fire when block-1 entropy $H_n < \tau_n$ (router-confidence rule).
- **B2.c.** Dynamic, fire when $w_{\text{recent},n} - w_{\text{embed},n} > \tau_n$ (relative-recent rule).

All three thresholds are calibrated as the q=0.5 per-position quantile on 32×8192 FineWeb-Edu tokens (§C.5). Results in Tab. 11; observed skip rates on the eval distribution in Tab. 12.

Conclusion. At a fair rate-matched comparison ($\sim$12% fired skips), *input-dependent* dynamic skip (B2.c) preserves LAMBADA at 0.345 while every static or random alternative collapses to 0.22–0.26 (-8 to -13pp). The same ordering holds on HellaSwag, PIQA, ARC-easy and OpenBookQA. The MoD-style “you might be getting a free lunch from any same-rate schedule” critique is rejected: at 340M from-scratch, schedule choice matters and the ATTNRES routing weights carry the load.

Table 11: **Decision-rule and threshold-rule ablation on** 340**M from-scratch** **ATTNRES**, lm-evaluation-harness with $\mathcal{P}=\{3,5\}$, $M_{\max}=1$. Dynamic rules whose threshold actually fires on the eval distribution (B2.c) preserve LAMBADA at 0.345 vs. static / random at the same $\sim$12% rate (0.22–0.26). Dynamic rules whose calibrated thresholds rarely cross on lm-eval data (B1.b at 3.1% observed; B2.b at 0.0%, see Tab. 12) are nearly equivalent to no-skip and are not the rate-matched comparison.

Cell	LAMBADA	HellaSwag	PIQA	ARC-e	OpenBookQA
B0. no-skip (upper bound)	0.4054	0.4607	0.6893	0.5438	0.3580
Dynamic, calibrated q=0.5 *on FineWeb-Edu*					
B1.b. recent_weight_gt	0.4011	0.4534	0.6839	0.5412	0.3580
B2.b. entropy_lt	0.4036	0.4608	0.6839	0.5417	0.3580
B2.c. recent_minus_embed_gt	**0.3445**	**0.4471**	**0.6774**	**0.5391**	**0.3540**
Static / random, rate-matched at 12.5%					
B1.c. static skip P=3 every	0.2624	0.3968	0.6610	0.5206	0.3300
B1.d. static skip P=5 every	0.2189	0.4223	0.6638	0.5253	0.3260
B1.e/B2.d. random $\{P$=3 OR P=5$\}$	0.2416	0.4028	0.6420	0.5101	0.3300

Table 12: **Observed skip rate on** 8×512 **LAMBADA tokens.** The calibration set (FineWeb-Edu) and the eval set (LAMBADA) have different routing-statistic distributions; B2.b's `entropy_lt` threshold never fires on LAMBADA and B1.b fires only 3.1%, so their headline-table accuracy is misleadingly close to no-skip. B2.c is the strategy whose calibration target ($\sim$12.5%) actually transfers, and is therefore the load-bearing rate-matched comparison against B1.c/d/e in Tab. 11.

Cell	Calibration target	Observed (LAMBADA)	Notes
B0	0%	0.00%	sanity check, $\tau_n=\infty$
B1.b	$\leq$ 12.5%	3.12%	τ_3=0.4645, τ_5=0.4010
B2.b	$\leq$ 12.5%	0.00%	τ_3=0.8631, τ_5=0.8764
B2.c	$\leq$ 12.5%	**10.94%**	τ_3=0.1655, τ_5=0.2472
B1.c	12.5%	12.50%	forced, every-token P=3
B1.d	12.5%	12.50%	forced, every-token P=5
B1.e	12.5%	per-batch toggle	uniform $\{P$=3, P=5$\}$

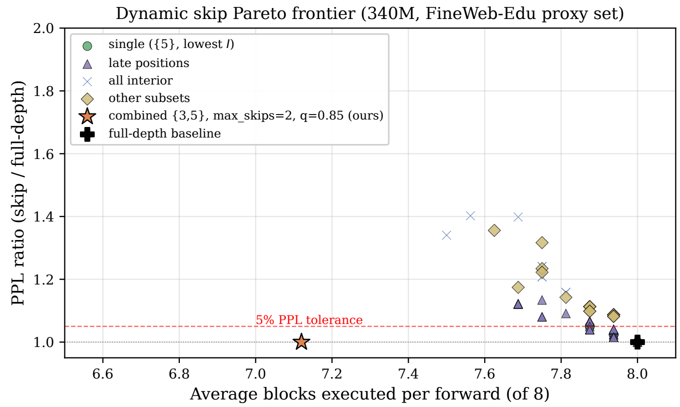
Threshold-transfer caveat. Two of the three dynamic rules (B1.b `recent_weight_gt` and B2.b `entropy_lt`) fire far below the 12.5% FineWeb-Edu calibration target on the LAMBADA distribution (Tab. 12). Their near-no-skip accuracy is therefore not a positive datapoint for those specific rules; it is consistent with "the threshold protected the output by accidentally not firing." The load-bearing rate-matched comparison is B2.c `recent_minus_embed_gt` (which does fire close to target) versus the static/random alternatives. We treat the two non-firing rows as a sensitivity result: RESKIP's safety margin under threshold/distribution mismatch is high (no degradation when the rule rarely fires), but deployment requires per-distribution threshold calibration, as already noted for the cross-modality VLA case (§F.4).

D.6 RESKIP full Pareto and latency curves at 340M

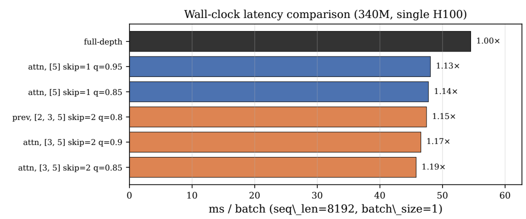
E Retrofit Pareto and breakdowns

E.1 LoRA baselines

We compare against parameter-matched LoRA baselines on the same $50/50$ UltraChat + LLaVA mix at $2\times$ the canonical retrofit's step count. LoRA r=32 on q, v: LAMBADA acc 0.540 / 0.516 (two seeds), HellaSwag 0.510 / 0.524. LoRA r=16 on q, k, v, o: LAMBADA 0.534, HellaSwag 0.492. LoRA r=8 on MLP: LAMBADA 0.514, HellaSwag 0.510. Mean LAMBADA across the four runs is 0.526, -0.6pp below base (0.532) and -5.0pp below our



(a) Accuracy–compute Pareto. $\mathcal{P}=\{3,5\}$ (orange star) sits below the 5% PPL tolerance.



(b) Wall-clock at seq 8192. $\{3,5\}/M{=}2/q{=}0.85$ reaches $1.19\times$.

Figure 3: ReSkip on 340M FineWeb-Edu: position-calibrated dynamic skip with $\mathcal{P}=\{3,5\}$ strictly dominates the single-position baseline at zero benchmark drop.

$\gamma\to 1$ retrofit (0.576). Mean HellaSwag 0.509 is +0.3pp over base versus retrofit's +1.6pp. Attribution: the retrofit's gain comes from the AttnRes routing structure, not from a few extra trainable parameters on SFT data.

E.2 Static-pruning baseline at the VLM scale (Gromov drop-k)

The 340M decision-rule ablation (Tab. 11) shows that *single-position* static skip and random skip both collapse on language tasks. The complementary question at the VLM scale is whether *full-layer* static pruning—the standard "remove the least useful blocks" baseline used by Gromov et al. [16]—can match the accuracy floor that the retrofit operates above. We rank the 28 Qwen3-VL-2B layers by the Gromov angular-distance criterion (computed on a FineWeb-Edu calibration set with the LM-head only; no fine-tuning), remove the k lowest-distance layers, and re-evaluate on `lmms-eval`. Tab. 13 reports drop-4 (layers $\{23, 24, 25, 26\}$) and drop-8 (additionally $\{12, 13, 14, 22\}$).

Table 13: **Static layer pruning on Qwen3-VL-2B** (Gromov-style angular-distance ranking, no retraining), `lmms-eval` full splits. Companion to the 340M in-block decision-rule ablation in Tab. 11: spans the *full-layer* vs. *single-position-within-block* skip dimension. Pruning 4 of 28 layers (14%) costs −16pp AI2D, −6pp MMMU, −5pp MMStar, −65pp OCRBench; pruning 8 (29%) is catastrophic on every cell. The retrofit (Tab. 2) operates at 0 removed layers, 0pp loss, and additionally lifts the base; ReSkip on the retrofit at $q{=}0.85$ holds within 1pp on LAMBADA (Tab. 14).

Configuration	AI2D	MMMU	MMStar	OCRBench	RWQA
Qwen3-VL-2B (base, $L_{\mathrm{rm}}{=}0$)	0.736	0.414	0.536	0.772	0.648
Gromov drop-4 (layers $\{23, 24, 25, 26\}$)	0.5732	0.3556	0.4906	0.1240	0.3791
Gromov drop-8 (drop-4 $\cup\ \{12, 13, 14, 22\}$)	0.0683	0.2389	0.0190	0.0030	0.1451
+ AR-Retrofit v3 ($L{=}4$, 10k; Tab. 2)	**0.758**	**0.432**	**0.536**	**0.814**	**0.661**

The pattern matches the in-block 340M observation: any *unconditional* block removal (single-position static at 340M, full-layer Gromov at 2B) destroys benchmark accuracy at the VLM scale even at modest pruning fractions, while input-dependent skip on top of an AttnRes routing structure preserves it. We do not attempt to retrain the pruned model; the intent is a free static-baseline floor for the cost-quality trade-off, not a tuned competitor.*

*Pruned cells additionally exhibit POPE generation drift below the strict yes/no surface (drop-4 pope-acc 0.021, drop-8 0.500 via accidental all-"yes" decoding), so we drop POPE from the table.

E.3 ReSkip on the canonical retrofit: text Pareto and cross-modality consistency

On the canonical L=4 v3 10k retrofit, applied to LAMBADA-500 as the language-modality cross-check of the LIBERO Pareto (Tab. 6), ReSkip produces the same shape as on action: lossless at the conservative end, collapsing once q moves below the action distribution's mean. Same $\mathcal{P}$=$\{1, 4\}$, $M_{\max}$=2 as the 2B VLA cell.

Table 14: **Cross-modality ReSkip consistency.** LAMBADA-500 on the canonical L=4 v3 retrofit, $\mathcal{P}$=$\{1, 4\}$, M=2, τ calibrated on 32 held-out LAMBADA prefixes at quantile q. Same operating-point structure as the LIBERO 4-suite Pareto: near-lossless above the calibration distribution mean ($q \geq 0.85$); rapid collapse below, where the threshold instructs the router to skip blocks it is normally executing.

q	LAMBADA acc	LAMBADA ppl	Δ acc vs no-skip	avg. skips / forward
no-skip (M=0)	0.5700	4.526	—	0.00 / ≤ 7
0.85	0.5600	5.258	-1.0	0.19 / 2
0.50	0.4120	12.550	-15.8	1.06 / 2
0.30	0.3900	14.005	-18.0	1.17 / 2

The legacy v1 (5k, $\mathcal{P}$=$\{4, 6, 11\}$) Pareto on the H_r256_5k retrofit *lifted* LAMBADA above its full-path value at q=0.95, M=1 (0.593 vs. 0.576), but that retrofit is no longer canonical for any reported result; we list the v1 Pareto in Tab. 15 for completeness.

Table 15: (Legacy) Dynamic skip Pareto on the v1 H_r256_5k retrofit, $\mathcal{P}$=$\{4, 6, 11\}$. Retained for reproducibility of the earlier draft.

q	$M_{\max}$	LAMBADA acc	LAMBADA ppl	avg. skips / 3 eligible
0.50	1	0.5500	5.45	0.82
0.70	1	0.5600	5.24	0.64
0.85	1	0.5733	4.90	0.37
0.95	**1**	**0.5933**	**4.77**	**0.18**
0.50	2	0.4733	6.42	1.41
0.70	2	0.5300	5.62	0.97
0.85	2	0.5600	5.03	0.55
0.95	2	0.5800	4.84	0.26

E.4 Wall-clock latency: full table including legacy L=2 and VLA in-backbone

Tab. 4 in the main body reports the canonical L=4 result. Tab. 16 below shows the full set: the legacy L=2 partition (14 blocks at 2B) carries a 1.39× structural cost over stock Qwen3-VL-2B because each token pays 14 router calls, the VLA in-backbone forward (the same retrofit run inside the OFT trainer's backbone forward, §4) tracks the VLM retrofit within 1% at both partitions, and `torch.compile` closes the L=2 residual to 1.16× but is more dramatic on L=4 where the router count is halved. This table is the basis for the canonical-partition switch from L=2 to L=4.

The structural per-block router stack (`stack`→`RMSNorm`→`softmax`→`einsum` over N completed blocks) does not go away under cached decode: at T=1 it fires N times per decoded token. Adapter rank ablation (Appendix C.2) confirms the adapter is not the bottleneck ($\leq 2\%$); the router itself, when uncompiled, is the floor. `torch.compile` (rows 5–8) collapses the small router gemms into a captured graph and tunes the matmul kernels for retrofit's small shapes, removing $\sim$92% of the L=4 structural gap. The remaining 2.9% is the small overhead of 7 surviving router calls per token; closing it would require a Triton-fused router that issues a single kernel for the entire stack–→–einsum sequence (open future work).

Skip on top of compile. Eager-mode ReSkip contributes a further 5–9% when used on its own at q=0.99, but cannot currently be composed with `torch.compile` in a single forward: the dyn-skip rule requires an `.item()`

Table 16: **Forward-pass latency, full sweep** on $1{\times}$H100, bf16, batch 1, median of 20 runs after 5 warmup. Eager rows: stock-HF base vs. retrofit at the indicated L. Compile rows wrap both with `torch.compile`, `dynamic=False`; `reduce-overhead` captures CUDA graphs, `max-autotune` additionally tunes per-kernel matmuls. The VLA in-backbone row is the canonical $L{=}4$ retrofit loaded into the OFT trainer's backbone forward (a separate code path) — within 1% of the VLM retrofit cell, confirming the per-block router stack, not the VLA harness, is the structural cost. *Bold:* the canonical operating point of the paper.

Configuration	seq	mode	base (ms)	retrofit (ms)	retrofit / base
(1) $L{=}2$ retrofit (legacy)	1024	eager	15.42	21.25	$1.378\times$
(2) $L{=}2$ retrofit (legacy)	2048	eager	26.03	36.14	$1.388\times$
(3) $L{=}4$ retrofit (canonical)	1024	eager	14.91	16.66	$1.117\times$
(4) $L{=}4$ retrofit (canonical)	2048	eager	25.49	28.53	$1.119\times$
(5) $L{=}2$ retrofit	1024	`reduce-overhead`	9.31	10.78	$1.158\times$
(6) $L{=}2$ retrofit	2048	`reduce-overhead`	21.25	24.45	$1.151\times$
(7) $L{=}4$ retrofit	2048	`reduce-overhead`	21.31	22.50	$1.056\times$
(8) $L{=}4$ retrofit	**2048**	**`max-autotune`**	**21.81**	**22.43**	**$1.029\times$**
(9) VLA in-backbone $L{=}4$ retrofit	2048	eager	25.49	28.79	$1.130\times$

on a per-token threshold comparison, which forces a CPU sync and breaks the captured CUDA graph. Production deployments pick one mode per latency budget—compile for high-throughput batch decode, eager-with-skip for variable-batch interactive workloads. A static-graph variant of the skip rule (compile-safe top-k with a fixed routing decision per shape bucket) is a known direction.

E.5 Compile vs. eager: accuracy parity

The near-parity inference cost of §3 requires that `torch.compile` preserve the retrofit's accuracy under fixed shape. We verified two parity tests on the canonical $L{=}4$ v3 10k retrofit.

LAMBADA-500 accuracy parity. Running LAMBADA-500 with the retrofit wrapped in `torch.compile` using default mode and dynamic shapes: acc 0.5720 compiled vs. 0.5700 eager ($\Delta{=}{+}0.20$pp), ppl 4.534 vs. 4.526. Per-target argmax agreement against eager: 98.60% — the residual 1.4% are tokens where eager and compiled both fall within bf16 SDPA jitter and the rank-1/rank-2 logits flip.

Per-token logit parity. On real prompt tokens with `mode="reduce-overhead"` (the inference-time mode used for the speed table): per-token argmax agreement 98.51%, max $|\Delta\text{logits}| = 0.50$, RMSE 0.060 — same magnitude as the cache-on/cache-off SDPA jitter on the stock base (Appendix B.6). Compile preserves the retrofit's forward to within bf16 evaluation noise; the $1.029\times$ `base_compiled` number is at iso-accuracy.

E.6 Block partition sweep

E.7 MMStar subcategory breakdown

F VLA appendix

F.1 VLA seed variance on 2B

For the two 2B suites with multiple rollout seeds, we report seed-level numbers below; per-suite variance is within $\approx$ 1pp. `libero_spatial` and `libero_object` are single rollouts at both scales.

Table 17: Block partition sweep (v3 recipe, r=256, γ-curriculum). L is layers-per-block, N=L_{total}/L. Per-layer L=1 underperforms by 2–9pp; the plateau $L \in \{2, 4, 7\}$ on 2B reproduces Kimi Team et al. [24]'s $S \in \{2, 4, 8\}$. We recommend L=4 on both scales because it preserves quality while reducing router calls.

Scale	L	N	LAMBADA	ΔL	HellaSwag	MMBench	MMMU	MMStar	AI2D	OCRBench	RWQA
2B base	—	—	0.532	—	0.506	75.77	0.414	0.536	0.736	0.772	0.648
2B retrofit	1	28	0.5645	+3.3	0.490	76.20	0.388	0.499	0.743	0.809	0.652
2B retrofit	2	14	0.5755	+4.4	0.494	77.23	0.426	0.532	0.748	0.803	0.663
2B (rec.)	**4**	**7**	**0.5650**	**+3.3**	**0.500**	**78.87**	**0.432**	**0.536**	**0.758**	**0.814**	0.661
2B retrofit	7	4	0.5155	−1.7	0.492	77.49	0.427	0.530	0.756	0.808	0.656
4B base	—	—	0.576	—	0.562	83.33	0.490	0.624	0.819	0.819	0.715
4B retrofit	1	36	0.5575	−1.9	0.523	83.33	0.497	0.538	0.783	0.768	0.694
4B retrofit	2	18	–	–	–	84.28	0.523	0.587	0.816	0.813	0.708
4B (rec.)	**4**	**9**	**0.6625**	**+8.7**	**0.552**	**85.22**	0.521	**0.632**	**0.825**	**0.824**	**0.718**
4B retrofit	6	6	0.6540	+7.8	0.554	84.79	**0.531**	0.623	0.817	0.824	0.715

Table 18: MMStar subcategory breakdown for the v3 retrofit. The math, logical, and sci&tech triple is the "reasoning over images" cell where the retrofit shows the largest gain (+7.9pp math on 2B; +3.9pp on 4B). Perception cells (coarse / fine / instance) are at parity or a small regression, consistent with a router that lifts deliberate-reasoning paths rather than altering early perception.

Configuration	math	logical	sci&tech	coarse	fine	instance
Qwen3-VL-2B (base)	0.413	0.429	0.408	0.714	0.520	0.710
+ AR-RETROFIT v3 (2B, L=4)	**0.492**	**0.432**	0.353	0.734	0.505	0.683
Δ vs. 2B base	**+7.9**	+0.3	−5.5	+2.0	−1.5	−2.7
Qwen3-VL-4B (base)	0.549	0.626	0.465	0.788	0.611	0.705
+ AR-RETROFIT v3 (4B, L=4)	**0.588**	0.602	**0.467**	0.812	0.606	0.714
Δ vs. 4B base	**+3.9**	−2.4	+0.2	+2.4	−0.5	+0.9

F.2 VLA pipeline pitfall: unfrozen action-token embeddings

Two Qwen3-VL-4B base VLMs we tried produced qualitatively different Path B behaviour. Our first attempt used an "Instruct-Action" variant that adds 2,048 `<robot_action_*>` embeddings (initialised at the mean of the existing vocab, kept unfrozen). The OFT head we train does *not* predict those tokens, but the rows for them sit in the tied `embed_tokens` / `lm_head` matrix and receive gradient from the retrofit's distillation loss and from label smoothing. Path B under this dirty base reaches 94.55% 4-suite avg (Long-10 84.8%); Path B under the clean base—no extra tokens added—reaches 96.70% (Long-10 94.2%). A +9.4pp Long-10 swing is entirely explained by whether 2,048 unused rows in the shared matrix receive stray gradient. Use the clean base for retrofit-+-OFT; only add bespoke action vocabulary when the head actually predicts those tokens.

F.3 VLA landscape: open VLAs on standard benchmarks

F.4 VLA RESKIP setup: per-block drift and eligible-set selection

The RESKIP eligible set $\mathcal{P}$ for each scale is selected from a per-block ablation *on the trained Path B policy*, not on the VLM retrofit alone, because the VLA action stream activates a different routing distribution than the language stream. We measure per-block action-drift MSE on a 24-sample sim trajectory: the policy is run with no skip and with each block individually skipped; the per-step action MSE between the skipped and no-skip rollouts is the per-block "cost of removing this block on the action distribution". The two safest blocks per scale form $\mathcal{P}$.

2B Path B v2 30k per-block action-drift MSE (sorted ascending, in units $\times 10^{-3}$): block 1: 0.4; block 4: 34.0; block 0: 58.1; block 5: 87.0; remaining blocks >100. Selecting the two lowest: $\mathcal{P}_{\text{2B}} = \{1, 4\}$.

Table 19: Per-seed Qwen3-VL-2B numbers for the two repeat-rollout suites. The corresponding cells in Tab. 5 report the per-seed mean. Seed 1 is the initial run; seed 2 is a fresh environment-reseeded rollout of the same trained checkpoint.

Suite	Policy	Seed 1	Seed 2
`libero_goal`	Path 0 (30k)	97.6	97.4
`libero_goal`	Path B (30k)	97.4	98.6
`libero_10`	Path 0 (30k)	92.8	91.4
`libero_10`	Path B (30k)	92.6	90.6

Table 20: Open VLA landscape on standard benchmarks. LIBERO entries are success rates (%) per task suite. Following common practice, π_0 and Octo-Base LIBERO numbers are taken from OpenVLA-OFT [23], which re-ran all three policies under a uniform LIBERO protocol. Our rows use the 30k Path B numbers from Table 5.

Model	#Params	Spatial	Object	Goal	Long-10	ref.
Octo-Base	93M	78.9	85.7	84.6	51.1	[33, 23]
OpenVLA	7B	84.7	88.4	79.2	53.7	[22, 23]
TraceVLA	7B	84.6	85.2	75.1	54.1	[51]
SpatialVLA	4B	88.2	89.9	78.6	55.5	[37]
π_0	3B	**96.8**	**98.8**	95.8	85.2	[3, 23]
Qwen3-VL-2B + AR-Retrofit + OFT (*ours*)	2B	**97.8**	99.6	**98.6**	92.6	this work
Qwen3-VL-4B + AR-Retrofit + OFT (*ours*)	4B	94.6	**99.8**	**98.2**	**94.2**	this work

4B Path B 30k clean per-block action-drift MSE: block 1: 0.6; block 2: 11.0; block 0: 34.0; block 3: 63.0; remaining >120. Selecting: $\mathcal{P}_{4\text{B}} = \{1, 2\}$.

Cross-scale transfer of $\mathcal{P}$ fails. Naively transferring 2B's $\mathcal{P}$=$\{1, 4\}$ onto 4B catastrophically drops LIBERO-Spatial to 12.4% at q=0.85 — block-4's drift on 4B is roughly $10\times$ that of block-2, so the same eligible-set rule produces wildly different operating points across scales. $\mathcal{P}$ must be calibrated on each model's own sim distribution; this is the per-scale half of the modality-aware skip protocol.

Threshold calibration. Per-block thresholds τ_n are the q-th empirical quantile of $w_{\text{recent},n}$ over 31,286 (2B) / 31,257 (4B) sim-rollout records (one record per decoded action token across many trajectories). The Tab. 6 numbers use this protocol. A separate "Method A" that calibrates on retrofit pretokenized data (as if the VLA were just a long-context LM) accidentally lands at a conservative operating point that mimics q=0.99 on the action distribution; we use it as a sensitivity comparator in our internal ablations and recommend the action-distribution-calibrated thresholds for any deployment.

Cross-modality threshold transfer is not safe. The LAMBADA-calibrated thresholds at q=0.85 from Tab. 14, applied unchanged to the same Path B policy at LIBERO inference, drop LIBERO-Spatial from 99.5% (a L=4 $v3$ warm-start, partial eval) to 64%. The action distribution has more low-w_{recent} mass than the language distribution at the same blocks; the language q=0.85 corresponds to roughly the action q=0.50, which over-skips. This is the empirical hard motivation for per-modality, per-token-class calibration in deployments that mix modalities within a single rollout.

F.5 VLA planned follow-ups

Per-modality, per-token-class ReSkip. The Tab. 6 thresholds are calibrated on the action distribution as a whole; an obvious refinement is per-modality $(\mathcal{P}^{(m)}, \tau_n^{(m)}, M_{\max}^{(m)})$ within a single rollout (vision tokens, language tokens, action tokens). Working hypothesis: vision α concentrates on early blocks (allows aggressive late-block skip on perception steps); action α spreads to late blocks (restricts late-block skip on action prediction steps). The calibration harness (`retrofit/eval/calibrate_vla_thresholds.py`) is in place; the failed cross-modality transfer of Appendix F.4 is the empirical motivation. *Compile + skip composition.* A static-graph variant of the skip rule (compile-safe top-k with a fixed routing decision per shape bucket) would let the compiled fixed-shape mode and the eager skip mode coexist in a single forward and stack their $1.029\times$ and 5–9% savings. *Edge hardware.* Wall-clock is H100 / bf16; RTX 4090 / Jetson Orin direct measurement is planned.

G NeurIPS Paper Checklist

1. **Claims.** Yes. The abstract, introduction, and conclusions state the main claims and report per-seed VLA numbers in the appendix where repeated rollouts exist.
2. **Limitations.** Yes. Appendix A lists architecture coverage, compile / dynamic-skip composition, hardware scope, and per-modality calibration limits.
3. **Theory assumptions.** Not applicable. The paper is empirical and algorithmic; all equations define implemented objectives or routing rules.
4. **Experimental reproducibility.** Yes. Sections 2.2–4.1 and the appendix specify model scales, block partitions, training steps, datasets, calibration rules, and evaluation protocols.
5. **Open access to code and data.** Partially. Experiments use public datasets and benchmarks where available; release details for retrofit / VLA training scripts will follow the anonymous-submission policy.
6. **Compute.** Yes. Main text and appendices report GPU type, training steps, rough GPU-minutes / GPU-hours, and latency measurement settings.
7. **Dataset and benchmark provenance.** Yes. All public datasets and benchmarks are cited; LIBERO and lmms-eval protocols are described with rollout counts or split usage.
8. **Human subjects.** Not applicable. No new human-subject data are collected.
9. **Privacy.** Not applicable beyond the privacy considerations of the cited public datasets.
10. **Licenses.** Partially. Public resources are cited; final camera-ready release will include the exact code / model / data license table.
11. **Broader impacts.** Yes. The work targets lower inference cost and adaptive computation in pretrained models; no safety-critical deployment is claimed.
12. **Safeguards.** Yes. VLA thresholds are calibrated per model and action distribution; cross-modality transfer failures are explicitly documented as a deployment risk.